

Special Topics in Discrete Mathematics: Advanced Topics in Cryptography

Theory Exercise 2

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April 2026



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^{*}Writing was done using $\text{\LaTeX}$

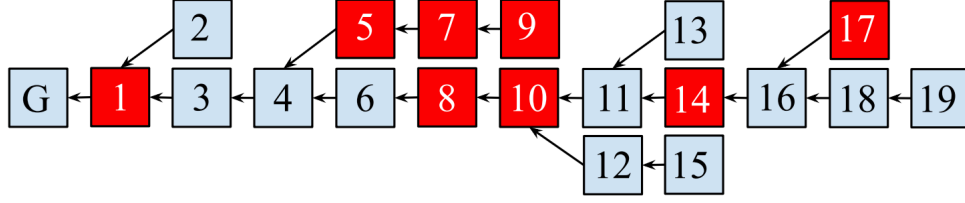
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Problem 1

Description

Consider the block tree in the figure below. Blue blocks are honestly mined blocks, whereas red blocks are adversarially mined blocks. Hypothetical block IDs are shown within the squares.



Let $\mathcal{C}_1, \mathcal{C}_2$ indicate the chains whose tips are blocks 9 and 19 respectively.

1. What is the chain quality of $\mathcal{C}_1[-4 :]$, $\mathcal{C}_1[: 4]$ and $\mathcal{C}_2$ respectively?
2. Suppose an honest party had adopted $\mathcal{C}_1$ at time 100 and $\mathcal{C}_2$ at time 200. What is the velocity τ of the chain between those two times?

Solution

1. We have

$$\mathcal{C}_1 = (G, 1, 3, 4, 5, 7, 9) \quad \text{and} \quad \mathcal{C}_2 = (G, 1, 3, 4, 6, 8, 10, 11, 14, 16, 18, 19).$$

By Definition 25, the chain quality of a chunk $\mathcal{C}[i : j]$ is

$$\mu = \frac{z}{|\mathcal{C}[i : j]|},$$

where z is the number of honest blocks in $\mathcal{C}[i : j]$.

For $\mathcal{C}_1[-4 :]$, the last four blocks are

$$(4, 5, 7, 9).$$

Only block 4 is honest, so

$$\mu(\mathcal{C}_1[-4 :]) = \frac{1}{4}.$$

For $\mathcal{C}_1[: 4]$, the first four blocks are

$$(G, 1, 3, 4).$$

The honest blocks are $G, 3, 4$, so

$$\mu(\mathcal{C}_1[: 4]) = \frac{3}{4}.$$

For $\mathcal{C}_2$, the honest blocks are

$$G, 3, 4, 6, 11, 16, 18, 19,$$

so there are 8 honest blocks out of 12 blocks in total. Hence

$$\mu(\mathcal{C}_2) = \frac{8}{12} = \frac{2}{3}.$$

Therefore,

$$\mu(\mathcal{C}_1[-4 :]) = \frac{1}{4}, \quad \mu(\mathcal{C}_1[: 4]) = \frac{3}{4}, \quad \mu(\mathcal{C}_2) = \frac{2}{3}.$$

2. By Definition 24, the chain velocity between times $r_1 < r_2$ is

$$\tau = \frac{|\mathcal{C}_2| - |\mathcal{C}_1|}{|r_2 - r_1|}.$$

Here,

$$|\mathcal{C}_1| = 7, \quad |\mathcal{C}_2| = 12, \quad r_1 = 100, \quad r_2 = 200.$$

Therefore,

$$\tau = \frac{12 - 7}{200 - 100} = \frac{5}{100} = \frac{1}{20}.$$

Problem 2

Description

Consider $\Delta = 1, n = 5, t = 2$. Draw a timeline of successful queries that could have caused Figure 1 to appear. For each successful query, indicate:

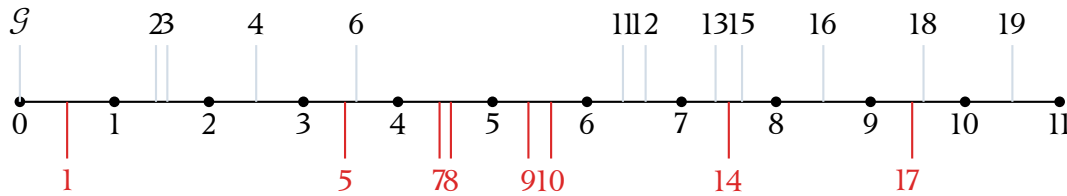
1. The time at which it took place.
2. Whether the query was honest or adversarial.
3. The time at which each honest party received the block produced by the query.

For the timeline you drew, what is the minimum $k \in \mathbb{N}$ for which Common Prefix holds between all honest parties and across all time.

Solution

One possible timeline of successful queries consistent with Figure 1 is the following. It is not unique, and the same timeline may also be compatible with other block tree figures.

Let the honest parties be H_1, H_2, H_3 (since $n = 5$ and $t = 2$, there are $n - t = 3$ honest parties).



- 1,2 The picture above gives the query times and indicates whether each successful query is honest (gray) or adversarial (red).
- 3 It remains to specify when each honest party receives each block.

We use the following convention: if an honest party mines a block at time s , then it receives it at time s , while the other two honest parties receive it at time $s + 0.3$. Since $0.3 < \Delta = 1$, this is consistent with the network delay bound.

We assign the honest blocks to the honest parties as follows:

$$H_1 : 2, 6, 13, 18,$$

$$H_2 : 3, 12, 15, 19,$$

$$H_3 : 4, 11, 16.$$

Then the reception times of the honest blocks are:

block	(H_1, H_2, H_3) receive times
2	(1.44, 1.74, 1.74)
3	(1.86, 1.56, 1.86)
4	(2.80, 2.80, 2.50)
6	(3.56, 3.86, 3.86)
11	(6.68, 6.68, 6.38)
12	(6.92, 6.62, 6.92)
13	(7.36, 7.66, 7.66)
15	(7.94, 7.64, 7.94)
16	(8.80, 8.80, 8.50)
18	(9.56, 9.86, 9.86)
19	(10.80, 10.50, 10.80)

For the adversarial blocks, one consistent choice is:

block	(H_1, H_2, H_3) receive times
1	(0.80, 0.80, 0.80)
5	(3.30, 3.30, 3.30)
7	(4.70, 5.70, 5.70)
8	(4.80, 4.80, 4.80)
9	(5.70, 5.70, 5.70)
10	(5.90, 5.90, 5.90)
14	(7.80, 7.80, 7.80)
17	(9.40, 9.40, 9.40)

These times are enough to explain the block tree. For example, 2 and 3 are mined within one delay window, so they can both extend block 1. Similarly, 11 and 12 can both extend block 10, and later the branches through 13 and 15 can coexist before all honest parties converge again.

For this timeline (that corresponds to the given figure), the minimum k for which Common Prefix holds is $k = 3$.

The reason is that at some time an honest party can adopt the chain

$$(\mathcal{G}, 1, 3, 4, 5, 7, 9),$$

while later all honest parties adopt

$$(\mathcal{G}, 1, 3, 4, 6, 8, 10, 11, 14, 16, 18, 19).$$

If we remove the last three blocks from the first chain, we get

$$(\mathcal{G}, 1, 3, 4),$$

which is a prefix of the later chain. But if we remove only the last two blocks, we get

$$(\mathcal{G}, 1, 3, 4, 5),$$

which is not a prefix of the later chain. Therefore $k \geq 3$.

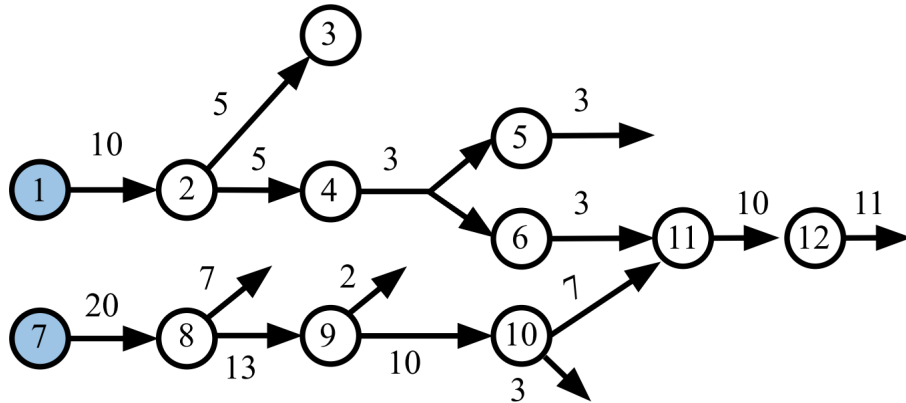
On the other hand, no fork in this timeline has a longer incompatible suffix than the branch ending at 9. Hence removing the last three blocks is always sufficient.

Therefore the minimum value is $k = 3$.

Problem 3

Description

Consider the UTXO transaction graph illustrated in the figure below. Hypothetical txids are shown within the circles. The value of an output is indicated above its respective arrow.



1. Honest party P has adopted a chain containing genesis (which has no transactions) and blocks B_1 (containing transaction 1 only) and B_2 (containing transaction 7 only) and is receiving the transactions from the network in this order: 2, 8, 9, 3, 4, 6, 5, 10, 11, 12. No other blocks beyond those three are mined. Which transactions will the mempool of this party contain?
2. The honest party managed to find a block B containing no new coinbase transactions and included its mempool on top of B_2 . The rest of the honest parties then mine another k sequential blocks on top of B . No other blocks are mined in the meantime. What is the ledger L^P reported by the honest party P at the end of this process?
3. How much monetary value remains unspent in the system in the view of party P ?

Solution

1. Initially, party P has adopted genesis, block B_1 containing transaction 1, and block B_2 containing transaction 7. Therefore the initial UTXO set consists of the output of 1 of value 10 and the output of 7 of value 20.

We now process the incoming transactions in the given order.

Transaction 2 is valid. It spends the output of 1 of value 10 and creates two outputs of value 5 and 5.

Transaction 8 is also valid. It spends the output of 7 of value 20 and creates two outputs of value 7 and 13.

Transaction 9 is valid as well. It spends the output of 8 of value 13 and creates outputs of values 2 and 10. Since

$$13 \geq 2 + 10,$$

the transaction is accepted.

Transaction 3 is valid (this is more of a convention - we could either enforce a dangling arrow with value zero or allow empty outputs). In the lecture notes the outputs are represented by a list, a fundamental list property is that it can be empty. Moreover, it spends one of the outputs of 2, namely the output of value 5, and creates no further output. This is still valid, because the total output value is 0, so the weak law of conservation is satisfied.

Transaction 4 is valid. It spends the other output of 2, namely the remaining output of value 5, and creates one output of value 3. Again,

$$5 \geq 3,$$

so it is accepted.

Transaction 6 is valid. It spends the output of 4 of value 3 and creates one output of value 3.

Transaction 5 is rejected. It tries to spend the same output of 4 that transaction 6 has already spent, so this outpoint is no longer in the current UTXO set.

Transaction 10 is valid. It spends the output of 9 of value 10 and creates outputs of value 7 and 3.

Transaction 11 is valid. It spends the output of 6 of value 3 and the output of 10 of value 7, so its total input value is 10, and it creates one output of value 10.

Transaction 12 is rejected. It spends the output of 11 of value 10, but it creates an output of value 11. Since

$$10 < 11,$$

this transaction is invalid.

Therefore the mempool of party P at the end is

$$(2, 8, 9, 3, 4, 6, 10, 11).$$

2. When P mines block B , it includes its mempool in order. Hence

$$B.x = (2, 8, 9, 3, 4, 6, 10, 11).$$

After that, the rest of the honest parties mine another k sequential blocks on top of B . At the end of the process, the adopted chain of P consists of genesis, B_1 , B_2 , B , and then these k additional blocks.

By the stable read rule, the ledger reported by P is obtained from the adopted chain after removing its last k blocks. Therefore the reported ledger contains exactly the transactions in genesis, B_1 , B_2 , and B . Since genesis contains no transactions, we get

$$L^P = (1, 7, 2, 8, 9, 3, 4, 6, 10, 11).$$

3. We now compute the monetary value that remains unspent in the system in the view of P . From part (2), the ledger is

$$L^P = (1, 7, 2, 8, 9, 3, 4, 6, 10, 11).$$

The initial value introduced by transactions 1 and 7 is $10 + 20 = 30$.

The unspent value is smaller than 30 for two reasons. First, transaction 3 spends an output of value 5 and creates no outputs, so this value is destroyed. Second, the transactions 4 and 9 pay fees: transaction 4 has fee $5 - 3 = 2$, and transaction 9 has fee $13 - (2 + 10) = 1$. In the notes, transaction fees are collected through the coinbase transaction of the block. Since the block B contains no coinbase transaction, these fees are not claimed and therefore are also removed from the system. Thus the total value removed is

$$5 + 2 + 1 = 8.$$

Hence the value that remains unspent is

$$30 - 8 = 22.$$

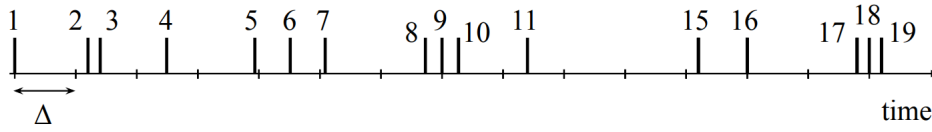
Equivalently, the final UTXO set consists of outputs of values 7, 2, 3, 10, whose sum is

$$7 + 2 + 3 + 10 = 22.$$

Problem 4

Description

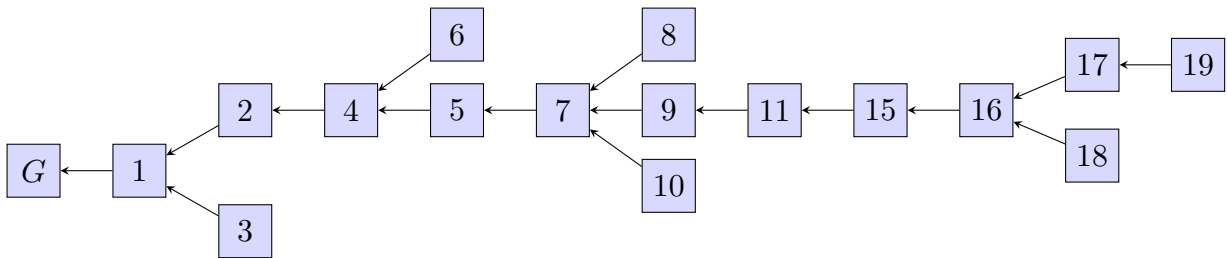
Consider the timeline of successful queries shown in the following figure. The network delay is $\Delta = 1$, and we have $n = 3$ and $t = 0$.



1. Indicate which among these successful queries are convergence opportunities.
2. Draw a blocktree that could have resulted from this timeline. For each block in the chain, indicate the successful query during which it was produced.
3. What is the height of the tip of the longest chain? What is the chain quality of the longest chain?

Solution

1. Since $t = 0$, all successful queries in the figure are honest. A successful query is a convergence opportunity if there is no other successful query within distance $\Delta = 1$ of it. From the timeline, the queries 2, 3 form one “group”, 5, 6, 7 form another, 8, 9, 10 form another, 15, 16 form another, and 17, 18, 19 form another. Hence none of these are convergence opportunities. The only “isolated” successful queries are 1, 4, and 11. Therefore the convergence opportunities are 1, 4, 11.
2. One possible blocktree consistent with the timeline is the following. It is not unique, since different blocktrees may arise from the same timeline depending on how the blocks propagate through the network.



Here block 1 is produced during successful query 1, block 2 during successful query 2, and so on.

3. In this blocktree, the longest chain is

$$(G, 1, 2, 4, 5, 7, 9, 11, 15, 16, 17, 19).$$

It contains 12 blocks including genesis, so its tip has height

$$12 - 1 = 11.$$

Since $t = 0$, all blocks are honest. Therefore every block in the longest chain is honest, and its chain quality is

$$\mu = \frac{12}{12} = 1.$$

Problem 5

Description

I was using the AI program ChatGPT to save some time while preparing the lecture notes for this course. As I was working on them, ChatGPT autocompleted my notes with the following text:

It seems that all three properties, collision resistance, preimage resistance, and second preimage resistance, are desirable. However, it is not possible to have all three at the same time. In fact, the following theorem shows that it is impossible to have collision resistance and second preimage resistance at the same time.

Theorem 1 (Krawczyk's Theorem). *Let $H : \{0, 1\}^* \rightarrow \{0, 1\}^\kappa$ be a hash function. Then, H is collision resistant if and only if it is not second preimage resistant.*

Prove or disprove the above theorem. You may use ChatGPT and all the theorems we have proven in class. Good luck!

Solution

We disprove the claim.

The statement

H is collision resistant if and only if H is not second preimage resistant

contains the two implications

H collision resistant $\implies H$ not second preimage resistant

and

H not second preimage resistant $\implies H$ collision resistant.

In the lecture notes, we proved

H collision resistant $\implies H$ second preimage resistant.

Hence the first implication is false.

Also, its contrapositive is

H not second preimage resistant $\implies H$ not collision resistant.

Hence the second implication is also false.

Therefore the stated theorem is false.